Electric Drives and Machines Institute
Graz University of Technology

Laboratory course
Control of Electric Drives and Machines
Course-Nr. 431.312

Unit 4:

Control of induction-machine, addendum

Teaching goals
Continuous time model

1 Parameters

From load tests around 1500 rpm the following parameters were identified. These allow for a simulation error of less than 5% in active power P_S ($200W < |P_S|$) and an error less than 3% in magnitude of stator current for these tests.

Parameter	Symbol	Unit	Value
Stator resistance	R_S	Ω	0.149
Rotor resistance	R'_R	Ω	0.103
Stator leakage inductance	$L_{\sigma S}$	H	$3.43e-4$
Rotor leakage inductance	$L'_{\sigma R}$	H	$5e-4$
Magnetising inductance	L_m	H	$8.27e-3$

Table 1: Proposed parameters for continuous time model at a constant rotor flux linkage of 0.08 Vs (amplitude). Converter voltage drop is not included.

These parameters might be used for continuous time model but not for the discrete time observer. The discrete time observer used in the lab was parameterised with parameters supplied by the attendees.

2 Continuous time model of induction machine

2.1 Overview

The following equations presume constant parameters.

- States: $\underline{\lambda}_S^S$ and $\underline{\lambda}_R^S$
- Input: Stator voltages v_a, v_b and v_c, mechanical rotor speed ω_{mech}
- Output: Stator currents i_a, i_b and i_c, inner torque T_{el} acting on the rotor.

Input quantities v_a, v_b and v_c are transformed into stator voltage space vector by applying matrix $\underline{T}$

$$\begin{bmatrix} v_{S\alpha} \\ v_{S\beta} \\ v_{S0} \end{bmatrix} = \frac{2}{3} \begin{bmatrix} 1 & -\frac{1}{2} & -\frac{1}{2} \\ 0 & \frac{\sqrt{3}}{2} & -\frac{\sqrt{3}}{2} \\ \frac{1}{2} & \frac{1}{2} & \frac{1}{2} \end{bmatrix} \cdot \begin{bmatrix} v_a \\ v_b \\ v_c \end{bmatrix} = \underline{T} \cdot \underline{v}_{abc} \tag{1}$$

Since the star point is not connected, zero sequence current is identical to zero. Therefore component v_{S0} is omitted in further considerations.

2.2 Voltage equations

With $\underline{v}_S^S = v_{s\alpha} + j \cdot v_{S\beta}$ stator voltage equation in stator reference frame is given by

$$\underline{v}_S^S = R_S \cdot \underline{i}_S^S + \dot{\underline{\lambda}}_S^S \tag{2}$$

With all rotor quantities referred to stator winding (dashes are omitted), rotor voltage equation for squirrel cage machine in stator reference frame is given by

$$0 = R_R \cdot \underline{i}_R^S + \dot{\underline{\lambda}}_R^S - j \cdot \omega \cdot \underline{\lambda}_R^S \tag{3}$$

with electrical rotor speed $\omega = \omega_{mech} \cdot p$ (p...number of polepairs).
Since integrators can handle real vectors, representation of space vectors as real vectors is preferred, e.g.

$$\underline{v}_S^S = \begin{bmatrix} v_{S\alpha} \\ v_{S\beta} \end{bmatrix}$$

The rotation by 90° now has to be accomplished by application of rotational matrix $\underline{R}$

$$\underline{R}(\varphi) = \begin{bmatrix} \cos(\varphi) & -\sin(\varphi) \\ \sin(\varphi) & \cos(\varphi) \end{bmatrix}$$

For $\varphi = \pi/2$ this is

$$\underline{R}(\pi/2) = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$$

Rearrangement of voltage equations into an appropriate form for simulation leads to

$$\begin{aligned} \dot{\underline{\lambda}}_S^S &= -R_S \cdot \underline{i}_S^S + \underline{v}_S^S \\ \dot{\underline{\lambda}}_R^S &= -R_R \cdot \underline{i}_R^S + \omega \cdot \underline{R}(\pi/2) \cdot \underline{\lambda}_R^S \end{aligned} \tag{4}$$

2.3 Currents

To get the currents, flux linkage equations are exploited

$$\begin{aligned} \underline{\lambda}_S^S &= L_S \cdot \underline{i}_S^S + L_m \cdot \underline{i}_R^S \\ \underline{\lambda}_R^S &= L_m \cdot \underline{i}_S^S + L_R \cdot \underline{i}_R^S \end{aligned} \tag{5}$$

with $L_S = L_{\sigma S} + L_m$ and $L_R = L_{\sigma R} + L_m$.
The inverse of the L-matrix

$$\underline{L} = \begin{bmatrix} L_S & L_m \\ L_m & L_R \end{bmatrix}$$

is given by

$$\underline{L}^{-1} = \frac{1}{L_S \cdot L_R - L_m^2} \cdot \begin{bmatrix} L_R & -L_m \\ -L_m & L_S \end{bmatrix}$$

and so stator current is given by

$$\underline{i}_S^S = \frac{L_R}{L_S \cdot L_R - L_m^2} \cdot \underline{\lambda}_S^S - \frac{L_m}{L_S \cdot L_R - L_m^2} \cdot \underline{\lambda}_R^S \tag{6}$$

and for rotor current

$$\underline{i}_R^S = -\frac{L_m}{L_S \cdot L_R - L_m^2} \cdot \underline{\lambda}_S^S + \frac{L_S}{L_S \cdot L_R - L_m^2} \cdot \underline{\lambda}_R^S \tag{7}$$

holds.

2.4 Torque

$$T_{el} = \frac{3}{2} \cdot p \cdot \Im\left\{\underline{\lambda}_m^S \cdot \underline{i}_R^{S*}\right\} \tag{8}$$

The rotor is assumed to be magnetically smooth, i.e. rotor current cannot develop torque with it's own field. In equation (8) the flux linkage $L_m \cdot \underline{i}_R^S$ as part of $\underline{\lambda}_m^S$ does not contribute to torque. The product $L_m \cdot \underline{i}_R^S \cdot \underline{i}_R^{S*}$ is a real number. The same applies for the rotor leakage flux linkage and so also

$$T_{el} = \frac{3}{2} \cdot p \cdot \Im\left\{\underline{\lambda}_R^S \cdot \underline{i}_R^{S*}\right\} \tag{9}$$

holds.

2.5 Output

To calculate the phase currents the inverse of transformation matrix $\underline{T}$ is applied.

$$\begin{bmatrix} i_a \\ i_b \\ i_c \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ -\frac{1}{2} & \frac{\sqrt{3}}{2} \\ -\frac{1}{2} & \frac{\sqrt{3}}{2} \end{bmatrix} \cdot \begin{bmatrix} i_{S\alpha} \\ i_{S\beta} \end{bmatrix} \tag{10}$$

The third column is omitted since $i_{S0} = 0$ holds.